LAB EXPERIMENT RECORD

Set 20: Stock Analysis

Department	AIML
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Day	Tuesday

Aim

To visualize stock prices of three companies using a line chart, bar chart and table, and to build an interactive Tableau dashboard.

Dataset: Stock Prices

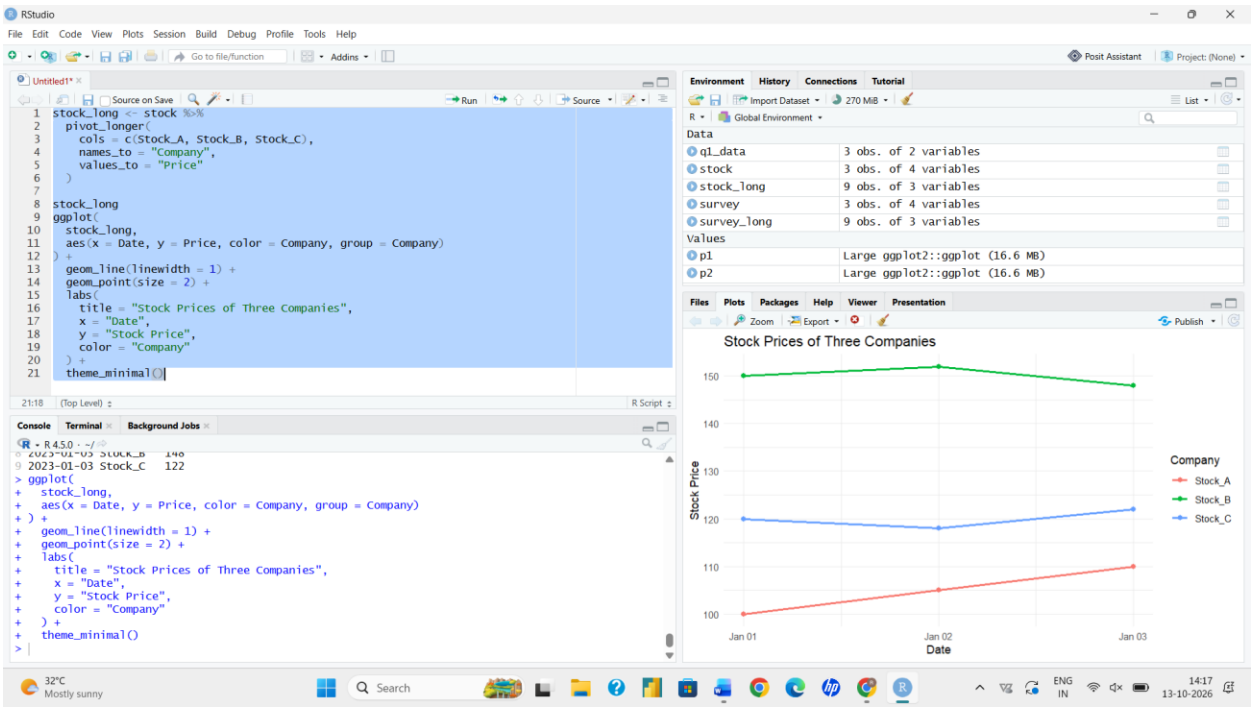
Date	Stock A	Stock B	Stock C
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

PROGRAM 1: Line Chart of Stock Prices

Create a line chart to visualize the stock prices of three companies (Stock A, Stock B and Stock C) over a specific time period. Label the axes and title the chart.

```
# Dataset
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03"))
stockA <- c(100, 105, 110)
stockB <- c(150, 152, 148)
stockC <- c(120, 118, 122)

# Line Chart
plot(date, stockB, type = "o", col = "green", ylim = c(95, 160),
     xlab = "Date", ylab = "Stock Price",
     main = "Stock Prices of Three Companies")
lines(date, stockA, type = "o", col = "red")
lines(date, stockC, type = "o", col = "blue")
legend("topright", legend = c("Stock A", "Stock B", "Stock C"),
       col = c("red", "green", "blue"), lty = 1, pch = 1)
```



Output 1: Stock Prices Line Chart

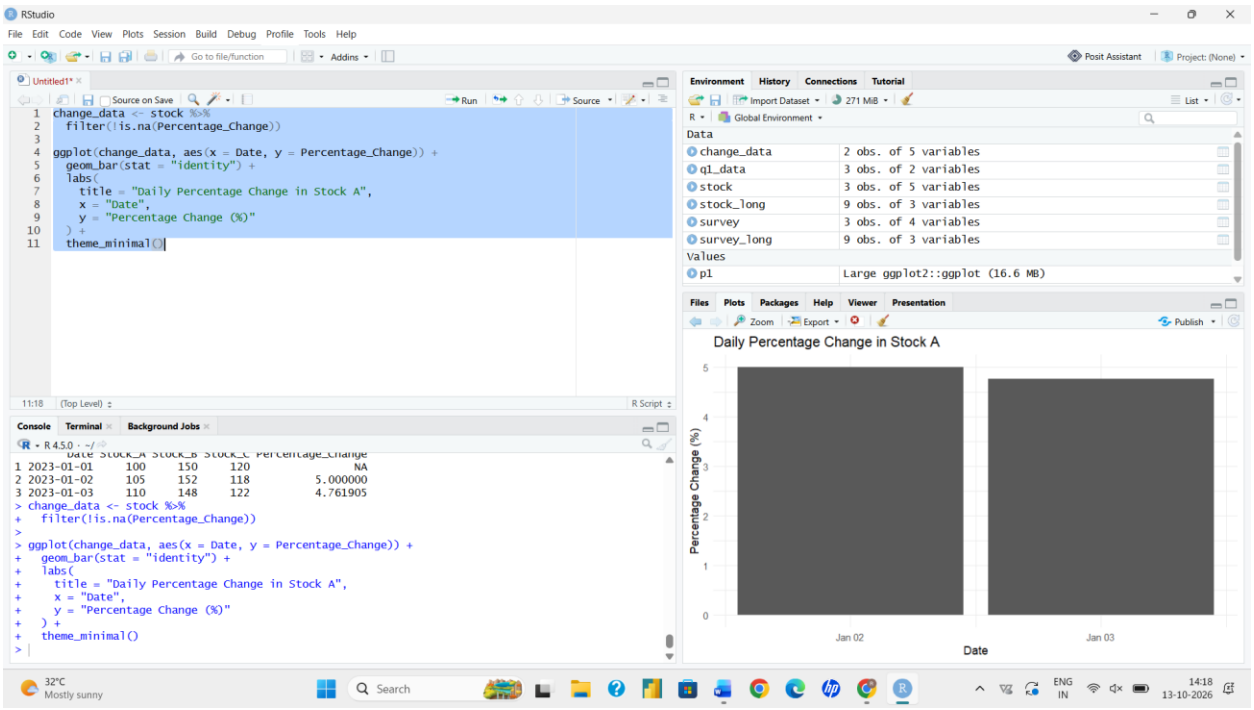
Observation: Stock A rises steadily from 100 to 110. Stock B stays the highest but falls to 148 on the last day. Stock C stays between 118 and 122.

PROGRAM 2: Bar Chart of Daily Percentage Change (Stock A)

Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.

```
# Daily percentage change for Stock A
pct_change <- diff(stockA) / head(stockA, -1) * 100
print(pct_change)           # 5.00  4.76

# Bar Chart
barplot(pct_change,
        names.arg = format(date[-1]),
        xlab = "Date",
        ylab = "Daily Change (%)",
        main = "Stock A: Daily Percentage Change",
        col = "steelblue")
```



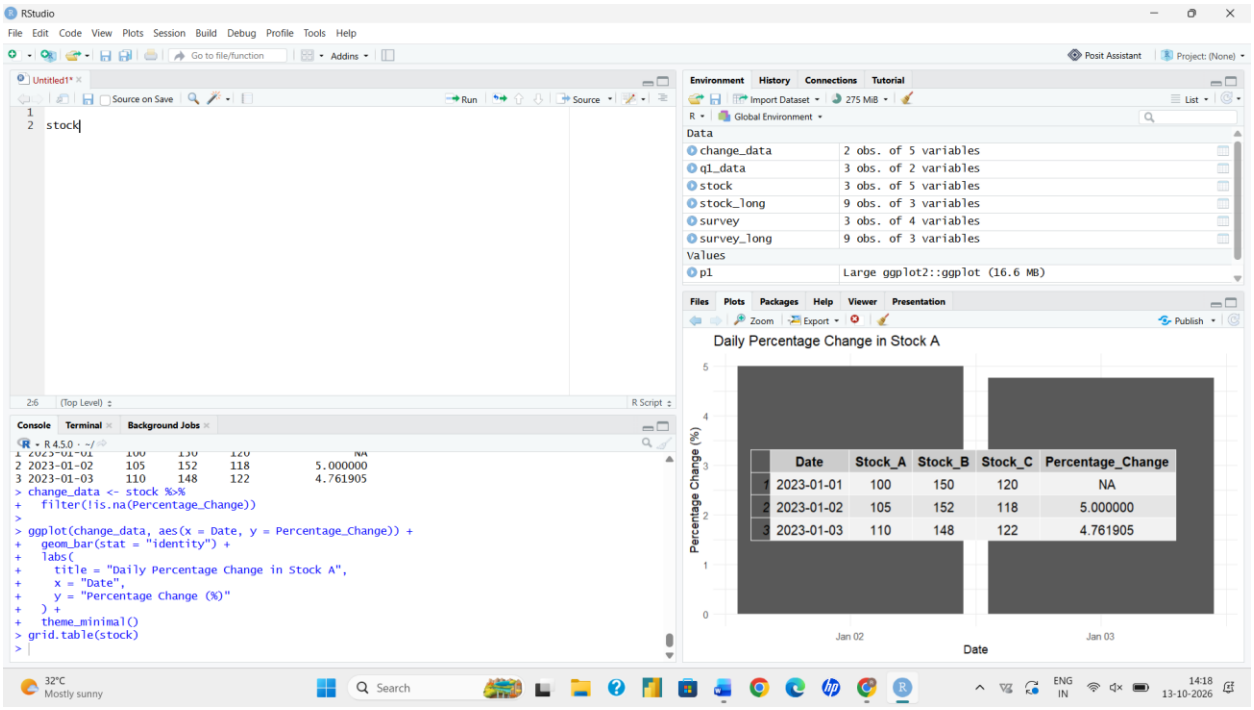
Output 2: Stock A Daily Percentage Change

Observation: Stock A increased by 5.00% on 2023-01-02 and by 4.76% on 2023-01-03. No percentage change can be calculated for 2023-01-01 because it is the first day.

PROGRAM 3: Stock Price Table

Build a table to display the stock price data for each company over the given period. Label the table elements.

```
stocks <- data.frame(Date = date,
                     StockA = stockA,
                     StockB = stockB,
                     StockC = stockC)
colnames(stocks) <- c("Date", "Stock A", "Stock B", "Stock C")
print(stocks)
knitr::kable(stocks, caption = "Stock Price Data")
```



Output 3: Stock Price Table

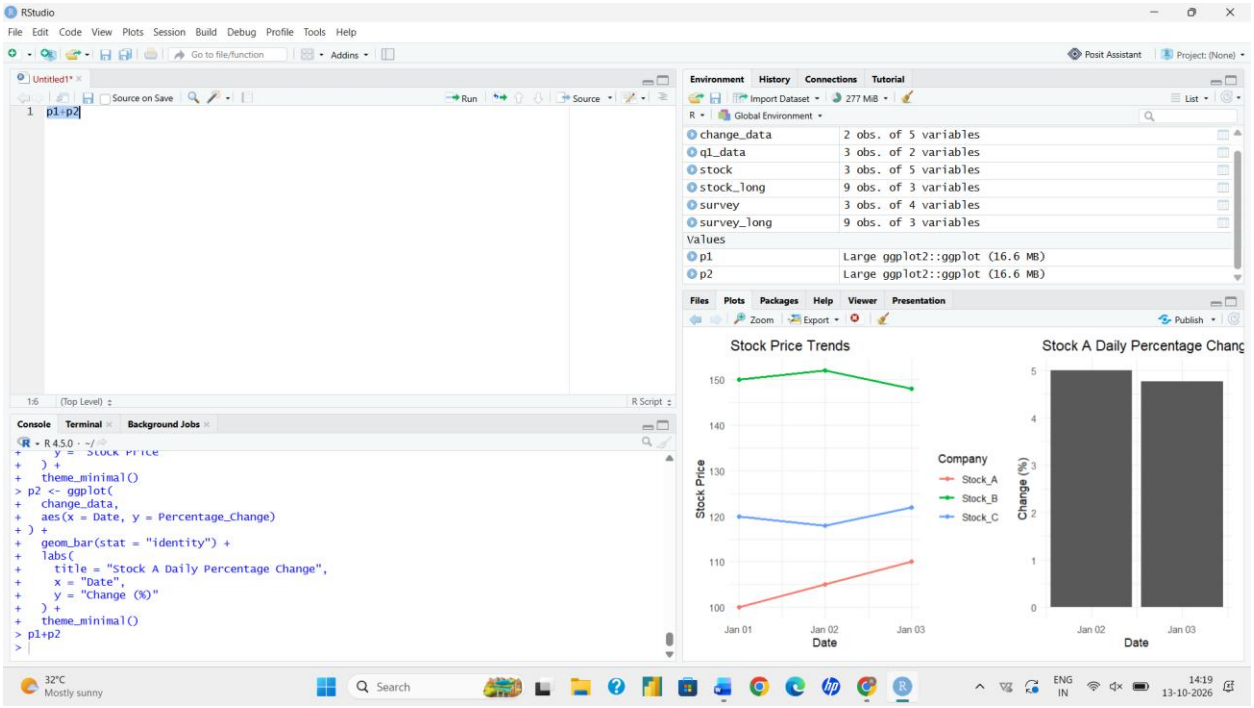
Observation: The table gives the closing price of each company for the three dates, with Date as the row label and each stock as a column.

PROGRAM 4: Interactive Tableau Dashboard

Develop a Tableau dashboard combining the line chart, bar chart and table for interactive exploration of stock prices.

Steps followed in Tableau:

1. Import the stock price file into Tableau and confirm that Date is recognised as a date field.
2. Select Stock A, Stock B and Stock C, and choose Pivot to create Stock and Price fields.
3. Sheet 1 (line chart): place Date on Columns, Price on Rows, and Stock on Color.
4. Sheet 2 (bar chart): filter to Stock A, then create a calculated field 'Daily % Change' using the Percent Difference quick table calculation on Price, and place it on Rows with Date on Columns.
5. Sheet 3 (table): place Date on Rows, Stock on Columns and Price on Text.
6. Create a Dashboard and drag the three sheets onto it. Add a Date filter and a Stock filter, and enable 'Use as Filter' on the line chart.
7. Add titles and axis labels, then save the workbook.



Output 4: Stock Prices Dashboard (layout of the Tableau dashboard)

Observation: The dashboard shows price trends, Stock A's daily change and the raw data together. The Date and Stock filters let the user focus on a particular company or period.

Procedure

1. Create date and stock price vectors in R.
2. Use plot() and lines() to draw the three-stock line chart with labelled axes and a legend.
3. Calculate Stock A's daily percentage change using diff() and plot it with barplot().
4. Combine the vectors in a data frame to display the stock price table.
5. Load the data in Tableau, build three sheets and combine them into an interactive dashboard.

Result

The stock prices were successfully visualized with a line chart, a bar chart and a table. A Tableau dashboard combined these views with filters for interactive exploration.

Key Functions Used

Function / Feature	Purpose
plot() / lines()	Draws and overlays line charts.
legend()	Adds a legend to the chart.
diff()	Calculates differences between consecutive values.
barplot()	Creates bar charts.
data.frame() / kable()	Builds and displays tables.
Tableau Dashboard	Combines sheets with filters.